Task 6: Sales Trend Analysis Using Aggregations

Objective

Analyze monthly revenue and order volume using SQL aggregation functions.

Tools Used

MySQL Workbench

Dataset

Sales Dataset containing Order Date, Amount, Customer, Product and Transaction details.

SQL Concepts Used

GROUP BY, SUM(), COUNT(DISTINCT), YEAR(), MONTH(), ORDER BY, Filtering.

Analysis Performed

- Monthly Revenue Analysis

```
select * FROM sales_dataset;
SELECT 'Order Date'
FROM sales_dataset
LIMIT 5;
select year(`order date`) as year, month(`Order Date`) as month, sum(amount)as total_revenue
from sales_dataset group by year(`order date`), month(`order date`)
order by year, month;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

year	month	total_revenue
2020	3	22991
2020	4	133385
2020	5	113287
2020	6	46900
2020	7	38556
2020	8	91117
2020	9	109434
2020	10	110836

Insight

Shows how revenue changes month by month.

- Monthly Order Volume Analysis

```sql
SELECT YEAR(`order date`) AS year, MONTH(`order date`) AS month,
    COUNT(DISTINCT `order id`) AS order_volume
FROM sales_dataset
GROUP BY YEAR(`order date`), MONTH(`order date`)
ORDER BY year, month;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

year	month	order_volume
2020	3	3
2020	4	16
2020	5	14
2020	6	9
2020	7	7
2020	8	13
2020	9	14
2020	10	16

Insight

Shows the number of orders received each month.

- Revenue and Order Volume Comparison

```sql
SELECT YEAR(`order date`) AS year, MONTH(`order date`) AS month,
    SUM(amount) AS total_revenue,
    COUNT(DISTINCT `order id`) AS order_volume
FROM sales_dataset
GROUP BY YEAR(`order date`), MONTH(`order date`)
ORDER BY year, month;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

year	month	total_revenue	order_volume
2020	3	22991	3
2020	4	133385	16
2020	5	113287	14
2020	6	46900	9
2020	7	38556	7
2020	8	91117	13
2020	9	109434	14
2020	10	110836	16

- Top Revenue Months Identification

```sql
SELECT YEAR(`order date`) AS year, MONTH(`order date`) AS month, SUM(amount) AS total_revenue
FROM sales_dataset
GROUP BY YEAR(`order date`), MONTH(`order date`)
ORDER BY total_revenue DESC
LIMIT 5;
```

year	month	total_revenue
2022	12	204413
2022	8	198965
2023	7	181554
2021	10	156211
2022	10	155508

-Specific Time Period

```sql
SELECT
    YEAR(`order date`) AS year,
    MONTH(`order date`) AS month,
    SUM(amount) AS total_revenue
FROM sales_dataset
WHERE YEAR(`order date`)=2024
GROUP BY YEAR(`order date`), MONTH(`order date`)
ORDER BY month;
```

year	month	total_revenue
2024	1	47645
2024	2	108414
2024	3	119216
2024	4	138052
2024	5	117494
2024	6	140745
2024	7	129944

Key Findings

• Revenue fluctuated across months, indicating seasonal demand patterns.

• Certain months generated significantly higher revenue than others.

• Order volume generally followed revenue trends.

• Peak-performing months contributed a substantial share of annual sales.

• Sales analysis can help identify periods for targeted marketing and inventory planning.

Challenges Faced

- Column names contained spaces (e.g., Order Date) requiring backticks.
- Date formatting and aggregation functions needed validation before analysis.

Conclusion

SQL aggregations successfully identified monthly revenue and order trends, helping transform transactional data into actionable business insights.